

Le Huu Huy

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EDUCATION

Sai Gon University <i>B.S. Computer Science</i>	Step.2023 - June 2027 <i>Current GPA: 3.21/4.0</i>
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SKILLS

Languages: Java, PHP, Python, JavaScript/TypeScript, C/C++, C#, HTML/CSS
Frameworks & Libraries: Spring Boot, Spring Data JPA, Hibernate, MapStruct, Laravel, React 19, FastAPI, LangChain
Security & IAM: Spring Security, FusionAuth, OAuth2 (Google Login), JWT
Databases & Caching: PostgreSQL, MySQL, Redis, ChromaDB, NoSQL
Architecture & Tools: Docker, Git/GitHub, Unix Shell, Maven, Postman, WebSocket, Cloudinary

PROJECTS

ShopZues — E-commerce Backend System *Spring Boot, MySQL, Redis, Firebase Realtime DB, FusionAuth, Docker* Nov 2025

- Built authentication flow with JWT (login, refresh, logout), including token invalidation via Redis blacklist and account verification states.
- Implemented RBAC with Role/Permission and method-level authorization to control admin vs. customer operations across APIs.
- Developed end-to-end order flow: create order, validate stock, compute shipping fee by package dimensions/weight, apply voucher/point discount, and persist price snapshots.
- Applied State Pattern for order status transitions (Pending → → Confirmed → → Packed → → Shipped → → Delivered/Completed/Cancelled/Refunded), reducing invalid state updates.
- Integrated VNPay callback verification (signature check + Redis TTL mapping) and synchronized order/payment status to Firebase Realtime DB for user/admin real-time tracking.
- Implemented OTP email verification and password-reset token flow with Redis TTL + FusionAuth; used Docker/Docker Compose for reproducible local environments (MySQL, Redis, FusionAuth).

<https://github.com/lhhuy02012005>

RAGAI (NexusRAG) — Hybrid Knowledge Base & Agentic Chat March 2026

- Tech Stack:** FastAPI, React 19, TypeScript, PostgreSQL, ChromaDB, LightRAG, Ollama, Sentence-Transformers, Docker
- Designed an end-to-end local-first RAG architecture: document upload → Docling parsing → chunking/dedup → embedding → vector + knowledge-graph indexing → hybrid retrieval → grounded generation.
- Implemented a hybrid retrieval pipeline combining ChromaDB vector search and LightRAG graph context, then improved precision with cross-encoder reranking (BAAI/bge-reranker-v2-m3).
- Built agentic chat with tool-calling (`search_documents`) and SSE streaming, delivering real-time events for thinking, tokens, sources, images, and final answer.
- Developed citation-first response format with source traceability (document/page/heading path), enabling explainable answers and quick navigation back to evidence.
- Added visual document intelligence: extracted images/tables from PDFs, generated captions, and injected media semantics into chunk embeddings for better retrieval quality.
- Engineered robust document lifecycle handling (PARSING/INDEXING/INDEXED/FAILED), timeout recovery, and async background processing for long-running ingestion tasks.
- Implemented workspace isolation across PostgreSQL metadata, vector collections, and knowledge-graph storage, supporting multi-tenant knowledge bases safely.
- Containerized infrastructure with Docker Compose (PostgreSQL + ChromaDB) and tuned runtime configs (chunk size, prefetch, rerank top-k, KG extraction limits) for both medium and high-spec machines.

- CryptionChat (ShieldTalk) — Secure End-to-End Encrypted Messaging PlatformMarch 2026
- **Tech Stack:** *Spring Boot 3, Java 17, Spring Security 6 (OAuth2 Resource Server), PostgreSQL, MongoDB, WebSocket (STOMP/SockJS), WebRTC, JWT (Nimbus JOSE), BCrypt, Web Crypto API (RSA-OAEP/AES-GCM/PBKDF2), Docker Compose*
 - Architected and delivered an **E2EE chat system** using a hybrid cryptographic model: **RSA-OAEP-2048** for key encapsulation and **AES-256-GCM** for message/file payload encryption, ensuring plaintext is only accessible on client devices.
 - Designed a secure key lifecycle with browser-side RSA keypair generation, **PIN-derived master key** via PBKDF2, encrypted private-key bundle storage (salt/iv/verification), and gated key unlock before protected routes are accessible.
 - Implemented a defense-oriented authentication architecture with **JWT (Nimbus JOSE)** over HttpOnly cookies, refresh-token flow, and **token versioning** to invalidate all prior sessions instantly after logout or security-triggered revocation.
 - Built a **hybrid data architecture**: PostgreSQL for identity, roles/permissions, and friendship graph; MongoDB for encrypted chat documents, attachment metadata, unread counters, and conversation retrieval at scale.
 - Developed realtime messaging infrastructure over authenticated STOMP channels for encrypted message delivery, read receipts, online presence, and friend-request status propagation.
 - Implemented **WebRTC video calling** with STOMP-based signaling events (RING/OFFER/ANSWER/BUSY/REJECT/END) and persistent call-history tracking for operational visibility and user activity timelines.
 - Engineered end-to-end social workflow (request/accept/decline) with realtime notifications, paginated friend listing/search, and computed friendship states to simplify frontend state transitions.
 - Hardened service boundaries using Spring Security resource-server configuration, cookie-based bearer-token resolution, and strict CORS policies for localhost and tunneled environments.
 - Containerized the development stack with Docker Compose (**PostgreSQL + MongoDB**) to ensure reproducible local environments, faster onboarding, and deployment parity.

EXPERIENCE

Sai Gon University

Ho Chi Minh City, Viet Nam

Software Engineering Student

Sept. 2023 – Present

- Developed high-performance web systems using **Laravel** and **Spring Boot**, focusing on Service-Repository patterns and clean architecture.
- Architected secure authentication systems using **JWT**, **OAuth 2.0**, and **Spring Security**, managing stateless sessions and role-based access control (RBAC).
- Optimized application performance by implementing **Redis** for distributed caching and asynchronous background tasks (queues), reducing database load during high-traffic simulations.
- Integrated third-party ecosystems including **VNPAY** for payment processing, **GHN** for logistics automation, and **Firebase** for real-time data synchronization.
- Containerized applications with **Docker** and orchestrated development environments using Docker Compose to ensure cross-platform consistency.
- Collaborated on version control via **Git/GitHub**, managing complex branching workflows and maintaining codebase integrity through code reviews.

HOBBIES

Music & Arts: **Playing badminton , Gym , Swimming.**

Tech Exploration: Researching **Generative AI (RAG)** and local LLM deployment; building custom mechanical keyboards.